

PerformaVis: Real-Time Affective Music Visualization Driven by Pianist's Bodily Expressions

Tai-Chen Tsai
Institute of Information Systems and
Applications
National Tsing Hua University
Hsinchu, Taiwan
astro5382@gapp.nthu.edu.tw

Chih-Chuan Huang
Interdisciplinary Program of
Management and Technology
National Tsing Hua University
Hsinchu, Taiwan
masonhuang0320@gmail.com

Shun-Han Chang
Department of Computer Science
National Tsing Hua University
Hsinchu, Taiwan
nthu11000224@gapp.nthu.edu.tw

Cheng-Yin Hsu
Department of Computer Science
National Tsing Hua University
Hsinchu, Taiwan
nelson32129@gmail.com

Hsin-Ying Lee
Interdisciplinary Program of
Management and Technology
National Tsing Hua University
Hsinchu, Taiwan
emily20030102@gapp.nthu.edu.tw

Kai-Hsiang Wen
Glisten Arts
Taipei, Taiwan
ken8338@gmail.com

Tse-Yu Pan
Institute of A.I. Cross-disciplinary
Tech
National Taiwan University of
Science and Technology
Taipei, Taiwan
typan@mail.ntust.edu.tw

Min-Chun Hu
Department of Computer Science
National Tsing Hua University
Hsinchu, Taiwan
anitahu@cs.nthu.edu.tw

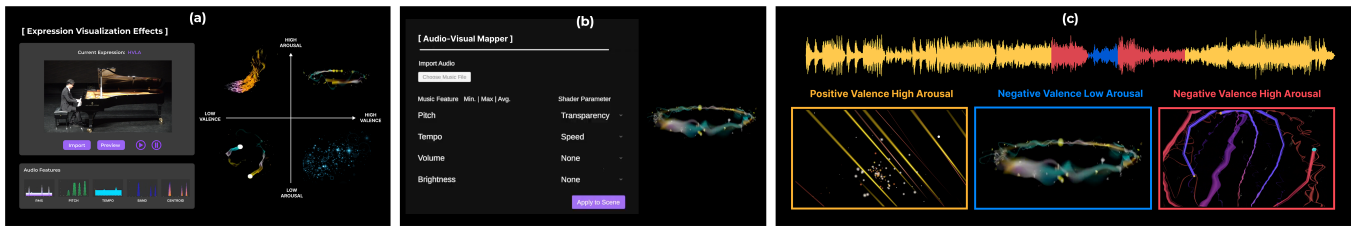


Figure 1: PerformaVis workflow. (a) The system supports streaming or video input and integrates an emotion recognition module to infer pianist's affective states during performance, transforming visual effects based on user-defined mappings. (b) Users define mappings between audio features and visual parameters. (c) The system adapts visual effects in real-time based on affective states and modulates fine-grained parameters according to audio features.

Abstract

Music visualization provides a cross-modal experience for interpreting abstract musical structures. However, existing systems overlook the affective interpretations conveyed by performers' bodily expressions, fundamental to live performance experience. We present PerformaVis, a system that combines bodily expression and audio features to generate personalized music visualizations. The system

infers affective states from body movements via emotion recognition and adapts visuals in real-time. To enhance personalization, we developed a multi-layered editor that enables customization of visual mappings for affect quadrants and audio features. Our pilot study shows that visualization driven by pianist's bodily expressions and audio features enhances audience affective engagement and interest more effectively than audio-driven visualizations.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758381>

CCS Concepts

• **Human-centered computing** → Visualization systems and tools; Interactive systems and tools.

Keywords

Music Visualization, Emotion Recognition, Performer Interaction

ACM Reference Format:

Tai-Chen Tsai, Chih-Chuan Huang, Shun-Han Chang, Cheng-Yin Hsu, Hsin-Ying Lee, Kai-Hsiang Wen, Tse-Yu Pan, and Min-Chun Hu. 2025. PerformaVis: Real-Time Affective Music Visualization Driven by Pianist's Bodily Expressions. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758381>

1 Introduction

Music visualization aims to enhance audience comprehension and immersion by transforming abstract auditory features into visual forms through cross-sensory translation. Early music visualization systems primarily focused on audio analysis, mapping acoustic features such as pitch, volume, rhythm, and energy to various forms of visual representation [4, 8, 12, 13]. Recent advances have introduced AI-driven methods into music visualization pipelines. Generative Disco [11] proposes an AI-driven music visualization pipeline that leverages LLMs and image generation models to create visuals for music segments, but requires users to manually segment audio and provide text prompts. In contrast, MusicJam [1] and Huang et al. [7] focus on automated approaches: MusicJam integrates a Multi-Modal Decoder with GPT-2 to automatically generate narrative descriptions and uses Stable Diffusion for corresponding image sequences, while Huang et al. developed a real-time system that extracts musical attributes (genre, instruments, emotions) via music information retrieval models, converts them to visual prompts using GPT-4 Turbo, and generates continuous abstract animations through StreamDiffusion. However, such audio-centric approaches overlook the affective interpretations conveyed by performers' bodily expressions, which are central to live musical performance [2, 3]. Different performers may interpret the same musical piece with varying affective nuances, yet most existing music visualization systems do not support these individualized expressions. While interactive systems have mapped performers' movements to visuals [5, 9], they rarely explore how bodily expressions convey affect for music visualization. To address these limitations, we introduce PerformaVis, a real-time affective music visualization system driven by pianist's bodily expressions. The system integrates an emotion recognition module and an audio feature extraction module. It continuously recognizes the pianist's affective states and categorizes them into 4 quadrants based on the affect circumplex model [14], with each quadrant mapped to specific visual effects through customizable configurations. Simultaneously, the audio feature extraction module analyzes musical characteristics to fine-tune visual parameters, ensuring comprehensive visual responsiveness to both emotional expression and musical content. We also developed a multi-layered visualization editor enabling customizable mappings between affective quadrants and visual effects (Figure 1a), as well as audio features and visual parameters (Figure 1b). The system combines real-time affect recognition with audio analysis, revealing both musical and emotional nuances in live performance.

2 Method

PerformaVis integrates emotion recognition, audio feature extraction, and visualization editing.

1. Emotion Recognition Module. We constructed a multi-modal piano performance dataset capturing synchronized audio, RGB-D video (top/side views), and wrist IMU data from five pianists. Performers self-annotated affective states using valence-arousal ratings, yielding 156 minutes of aligned multimodal recordings. We trained and evaluated emotion recognition models using two camera perspectives (top and side) and different input modalities (RGB and skeleton data). The TSM-based model [10] achieved optimal performance with RGB videos, effectively capturing subtle bodily expressions associated with affective states. While we explored skeleton-based approaches, occlusion from the piano body caused unstable keypoint detection, limiting their effectiveness in capturing complete motion patterns. We adopted the RGB model, which demonstrates practical performance (1.11s inference, 0.90 FPS, 1.9GB memory on RTX 3060), enabling dynamic response to performers' affective expressions for real-time visualization.

2. Audio Feature Extraction Module. To ensure visual output responds to both performer affective expression and acoustic qualities, we developed an Audio Feature Extraction Module analyzing time and frequency domain features. In the time domain, we extract tempo and RMS energy directly from the audio waveform. Tempo captures rhythmic pacing, while RMS energy measures overall loudness and intensity variations. In the frequency domain, Fourier transform enables extraction of pitch and spectral centroid. Pitch represents the fundamental frequency closely tied to emotional tone. Spectral centroid reflects timbral brightness, with higher values indicating sharper, high-frequency sounds and lower values for warmer tones. These features provide a foundation for personalized audio-visual mappings, enabling expressive visualization design.

3. Multi-layered Visualization Editor. Our multi-layered visualization editor enables user-defined mappings through two configuration layers. The first maps visual effects to four affective quadrants from the affect circumplex model, establishing direct affect-to-visual correspondence. The second defines custom relationships between audio features (tempo, energy, pitch, centroid) and visual parameters (brightness, saturation, particle density). The system includes research-based defaults for immediate use: tempo controls movement timing, energy modulates intensity, pitch affects hue, and spectral centroid adjusts brightness. Users can modify defaults or create new mappings. During operation with streamed or pre-recorded input, the Emotion Recognition Module determines affective quadrants for visual selection while Audio Feature Extraction continuously analyzes time/frequency features, synchronizing visual evolution with music, as shown in Figure 1(c).

3 Pilot Study

We conducted a pilot study with 14 participants (8M/6F, age=24.4) without professional music training, evaluating three conditions: Audio Only, Audio Visualization (audio features driven), and Emotional Audio Visualization (audio features and affect driven). Participants experienced randomized conditions and completed 5-point Likert questionnaires including NASA-TLX [6] and 8 experience dimensions. Results showed that the Audio condition achieved the highest focus score (mean = 4.43 ± 0.49), suggesting unimodal input reduces distractions. Both AudioVis and Affective-AudioVis demonstrated comparable ratings in meaningfulness, understanding, and

willingness to participate again, indicating visual enhancement increases experiential value compared to audio-only experiences. The Affective-AudioVis condition outperformed others in affective engagement (mean = 4.21 ± 0.45) and interest enhancement (mean = 3.71 ± 0.9). Immersion and task load remained comparable across conditions. These findings suggest that integrating performer-driven affective cues with audio features enhances audience engagement more effectively than audio-driven visualizations.

4 Discussion and Future Works

Our findings demonstrate that integrating performer-driven affective cues with audio features enhances audience engagement more effectively than audio-only visualizations. The system's accuracy is currently limited by scarce annotated affective body movement data. We plan to explore audio-movement multimodal fusion to improve emotion inference robustness under data-sparse conditions.

Acknowledgments

This research was funded by the National Science and Technology Council (NSTC) under grant numbers NSTC 114-2223-E-007-012-MY3 and NSTC 112-2221-E-007-079-MY3. We extend our sincere gratitude to all participants involved in the user research.

References

- [1] Chuer Chen, Nan Cao, Jiani Hou, Yi Guo, Yulei Zhang, and Yang Shi. 2023. MusicJam: Visualizing Music Insights via Generated Narrative Illustrations. *arXiv preprint arXiv:2308.11329* (2023).
- [2] Sofia Dahl and Anders Friberg. 2007. Visual perception of expressiveness in musicians' body movements. *Music Perception* 24, 5 (2007), 433–454.
- [3] Jane Davidson and Jorge Salgado Correia. 2002. Body movement. In *The science and psychology of music performance: Creative strategies for teaching and learning*. Oxford University Press, 237–249.
- [4] Sam Ferguson, Andrew Vande Moere, and Densil Cabrera. 2005. Seeing sound: Real-time sound visualisation in visual feedback loops used for training musicians. In *Ninth International Conference on Information Visualisation (IV'05)*. IEEE, 97–102.
- [5] Jia Fu, Jiarui Tan, Wenjie Yin, Sepideh Pashami, and Mårten Björkman. 2023. Component attention network for multimodal dance improvisation recognition. In *Proceedings of the 25th International Conference on Multimodal Interaction*. 114–118.
- [6] Sandra G Hart. 2006. NASA-task load index (NASA-TLX); 20 years later. In *Proceedings of the human factors and ergonomics society annual meeting*, Vol. 50. Sage publications Sage CA: Los Angeles, CA, 904–908.
- [7] Jenny Huang, Christoph Johannes Weber, and Sylvia Rothe. 2025. An AI-driven Music Visualization System for Generating Meaningful Audio-Responsive Visuals in Real-Time. In *Proceedings of the 2025 ACM International Conference on Interactive Media Experiences*. 258–274.
- [8] Nobuhiko Jin, Naoki Haga, and Issei Fujishiro. 2016. SeeGroove2: An Orbit Metaphor for Interactive Groove Visualization. In *2016 International Conference on Cyberworlds (CW)*. 131–134. <https://doi.org/10.1109/CW.2016.26>
- [9] Kyungho Lee, Donna J Cox, Guy E Garnett, and Michael J Junokas. 2015. Express it! An Interactive System for Visualizing Expressiveness of Conductor's Gestures. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition*. 141–150.
- [10] Ji Lin, Chuang Gan, and Song Han. 2019. Tsm: Temporal shift module for efficient video understanding. In *Proceedings of the IEEE/CVF international conference on computer vision*. 7083–7093.
- [11] Vivian Liu, Tao Long, Nathan Raw, and Lydia Chilton. 2023. Generative disco: Text-to-video generation for music visualization. *arXiv preprint arXiv:2304.08551* (2023).
- [12] Delfina Malandrino, Donato Pirozzi, Gianluca Zaccagnino, and Rocco Zaccagnino. 2015. A color-based visualization approach to understand harmonic structures of musical compositions. In *2015 19th International Conference on Information Visualisation*. IEEE, 56–61.
- [13] Dionysios Politis, Dimitrios Margounakis, and Konstantinos Mokos. 2004. Visualizing the chromatic index of music. In *Proceedings of the Fourth International Conference on Web Delivering of Music, 2004. EDELMUSIC 2004*. IEEE, 102–109.
- [14] James A Russell. 1980. A circumplex model of affect. *Journal of personality and social psychology* 39, 6 (1980), 1161.